Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
4	*(a)*			5 points plotted correctly to within <1 small square division (2) 4 correct (1) 3 or less (0) Straight line drawn [within the range of points] with a ruler through the points <1 small square division from plotted points, avoiding the anomalous point (1)		3		3	3	3
	(b)	(i)		Reading taken from candidate's graph (eg. 5.0 [N]) [<1 small square tolerance]		1		1	1	1
		(ii)		Use of a pair of complementary values, e.g. m = 5.0 **ecf** from graph ÷ 2.0 (1) [<1 small square tolerance] = 2.5 [kg] (1) NB. Use of (4.0 N, 2.6 m/s^2) only credited for either mark if the candidate's graph passes through the point.	1	1		2	2	2
		(iii)		Straight(ish) line with smaller positive gradient (by eye) drawn always below the original one. NB does not need to pass through the origin but cannot meet the existing graph [except at the origin].		1		1	1	1

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	(c)			Possible answers: 2×3.2 [m/s^2] = 6.4 [m/s^2] (1) ≠ 5.6 [m/s^2] so suggestion not correct (1) 16 [N]/ 5.6 [m/s^2] = 2.9 [kg] (1) ≠ 2.5 [kg] **ecf** from (b)(ii), so suggestion not correct 16 [N] / 2.5 [kg] (ecf from (b)(ii)) = 6.4 [m/s^2] (1) ≠ 5.6 [m/s^2] so suggestion not correct **Alternative:** Adding accelerations from forces adding up to 16 N, e.g. 4.8 + 0.8 + 0.8 = 6.4 (1) ≠ 5.6 [m/s^2] so suggestion not correct (1) NB. Use of (4.0 N, 2.6 m/s^2) not penalised in this part, e.g. from 12 N + 4 N 4.8 [m/s^2] + 2.6 [m/s^2] = 7.4 [m/s^2] (1) ≠ 5.6 [m/s^2] so suggestion not correct. (1) **Alternative:** Showing clearly that a different force, i.e. 14 N **ecf**, gives an acceleration of 5.6 m/s^2 (1) ≠ 16 [N] so suggestion not correct (1)			2	2		
				Question 4 total	**1**	**6**	**2**	**9**	**7**	**8**